



Batterjee Medical College
For Science and Technology

Thyroid Disorders

Internal Medicine Department

Knowledge

- Relate pathophysiological basics and aetiology of thyroid disorders
- Describe epidemiology, manifestations, complications and management of hyperthyroidism

Skills

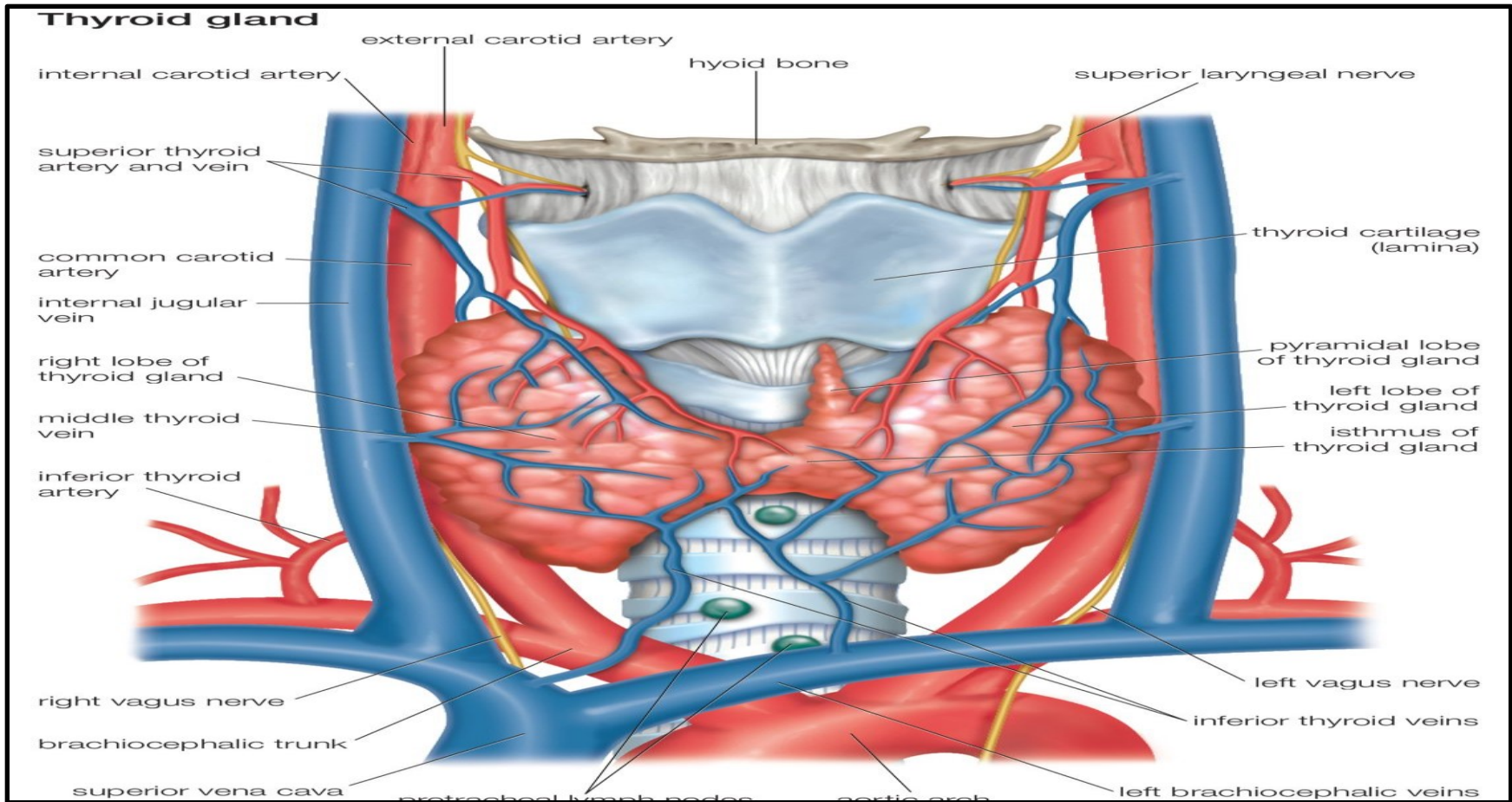
- Interpret clinical manifestations and investigations of hyperthyroidism to formulate reasonable diagnosis.
- Apply evidence-based management plan for hyperthyroidism
- Distinguish patients in emergency situations to formulate a comprehensive management plan; ensuring keeping patients in normal homeostasis

- Thyroid gland anatomy and physiology (hormone synthesis, hypothalamic-pituitary-thyroid axis, physiological effects of thyroid hormones)
- Thyroid function tests and antibody markers (TSH, T3/T4, anti-thyroid antibodies, thyroglobulin, radioactive iodine uptake)
- Hyperthyroidism and thyrotoxicosis (definition, causes, classification)
- Graves' disease (aetiology, clinical manifestations, thyrotoxic crisis, differential diagnosis)
- Management of hyperthyroidism (thionamides, beta-blockers, radioiodine therapy, surgery, treatment during pregnancy, treatment of complications)



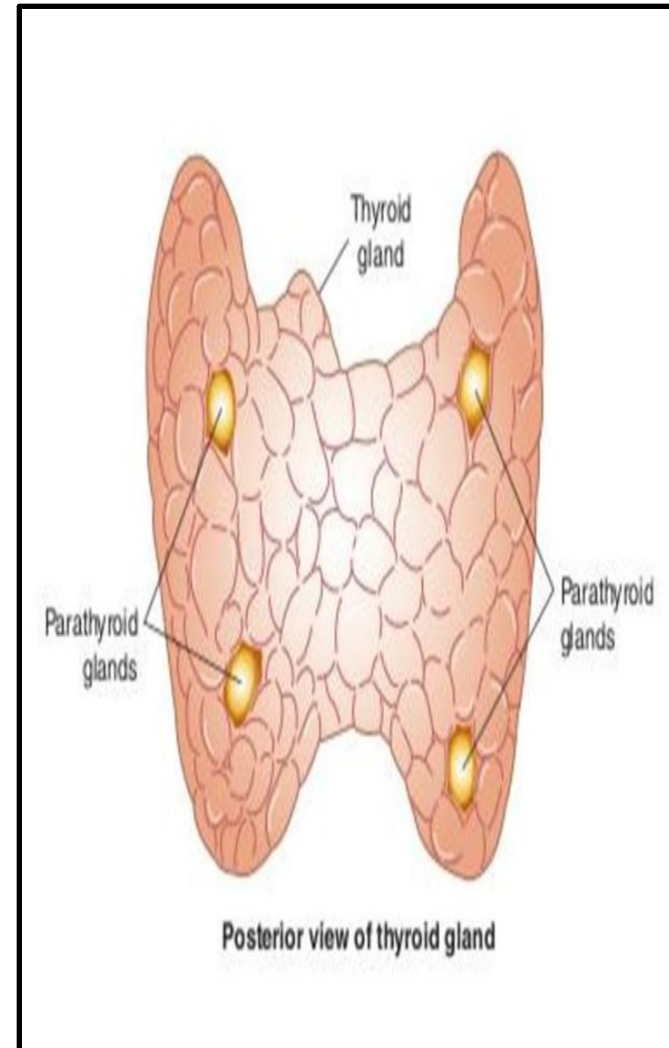
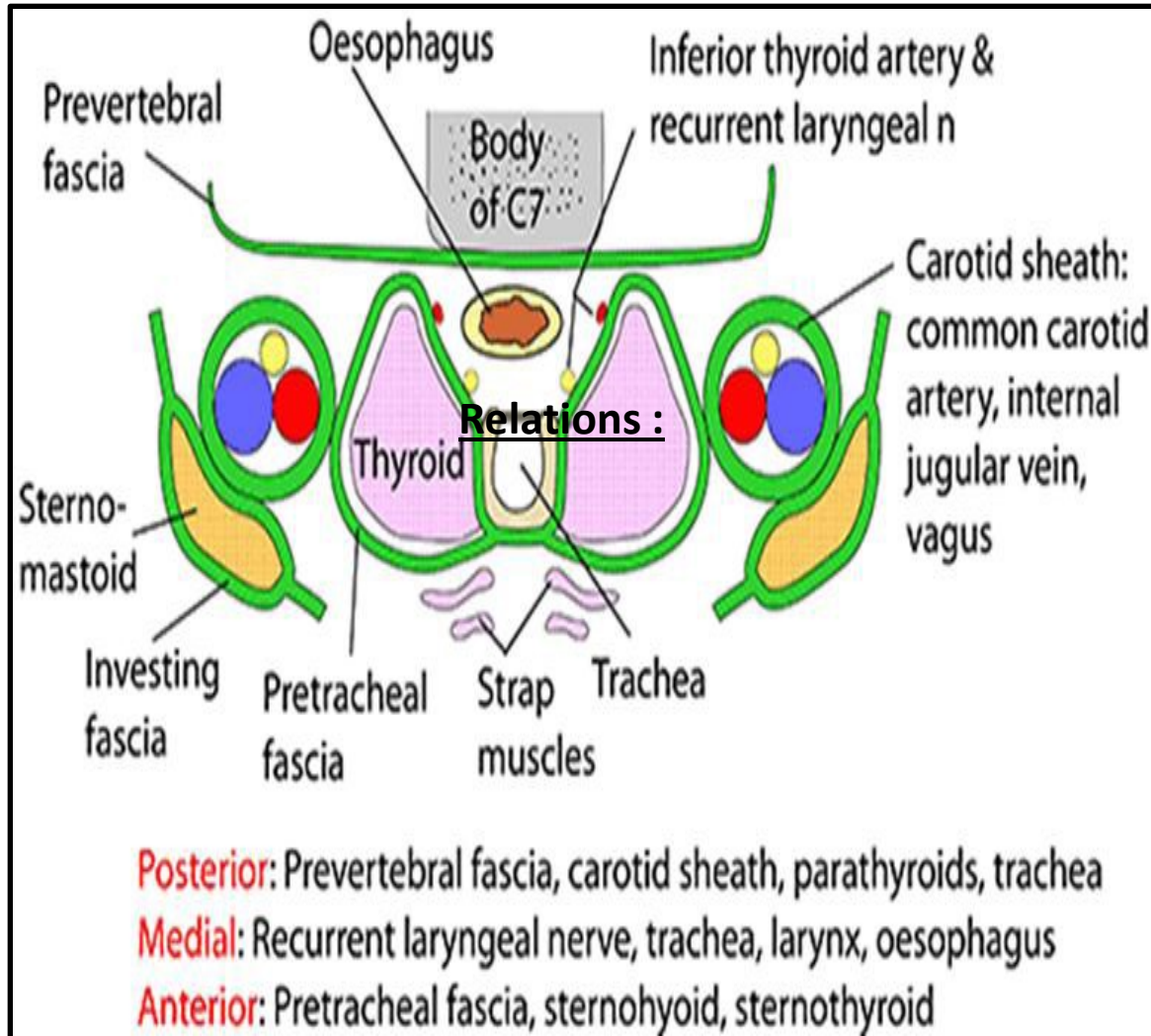
Thyroid Gland

Anatomy

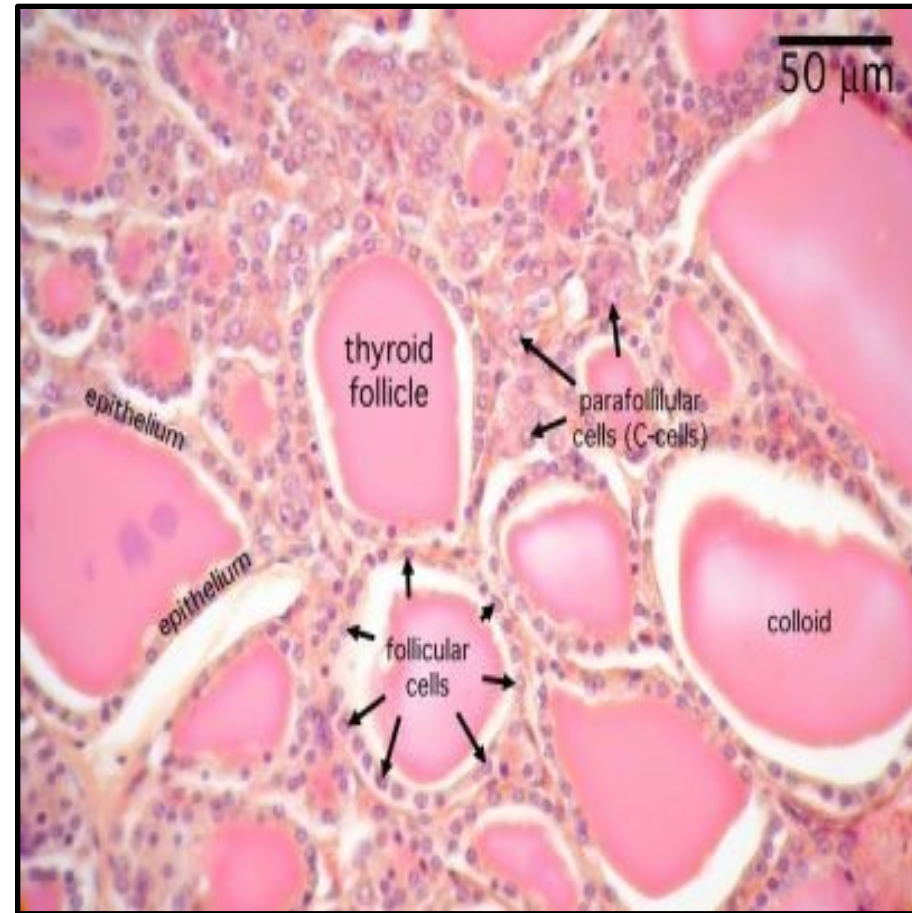
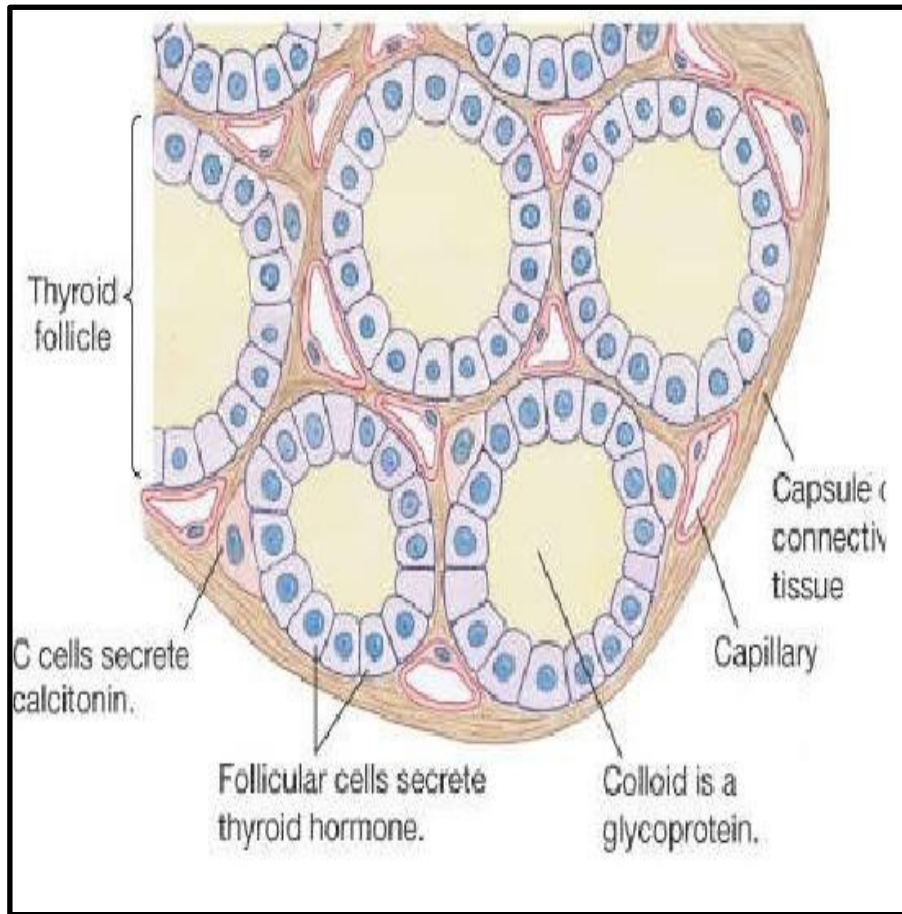


- Thyroid is one of the biggest endocrine glands, located in the neck anterior to the trachea, below the thyroid cartilage. Thyroid is butterfly shaped gland composed of two lobes connected by the isthmus.
- Weight about 12-20 g.

Relations



Histology



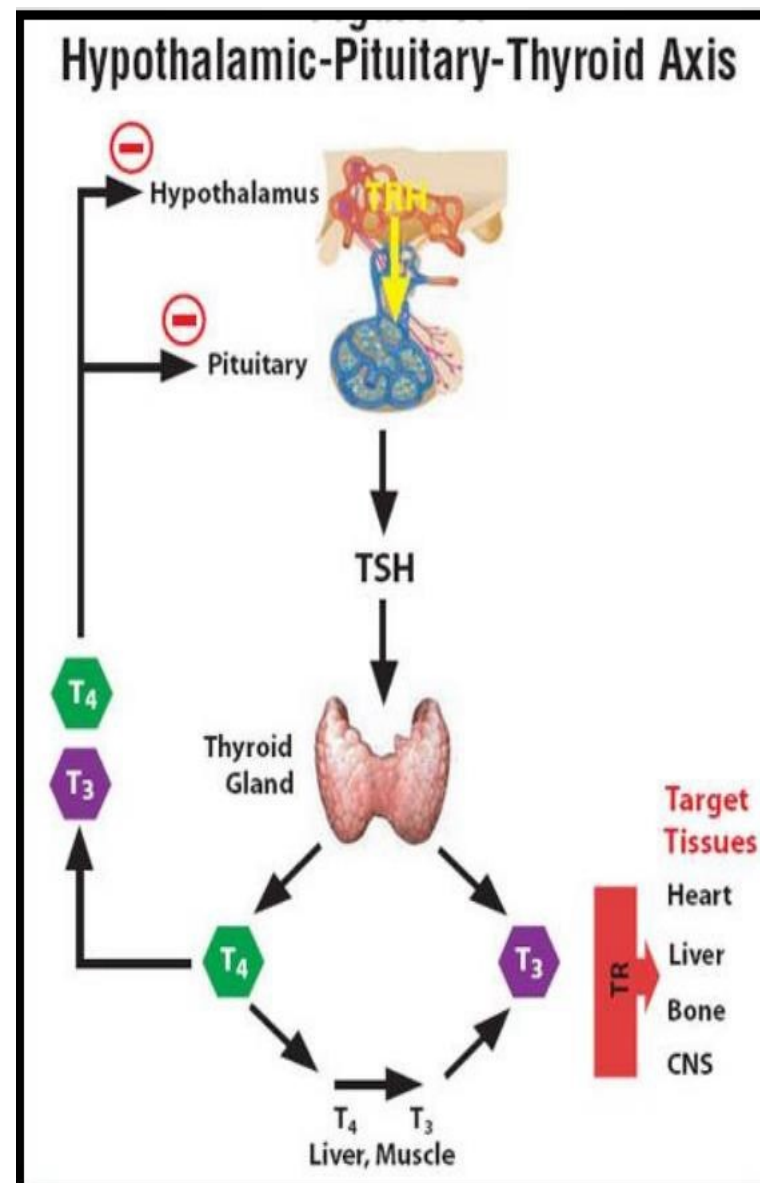
- Histologically consist of numerous follicles made from single layer of epithelial cells and contain colloid material rich in thyroglobulin, the precursor of T4 (Thyroxine) and T3 (Tri-iodothyronine).

Physiological effects of thyroid hormones

- ↑ O₂ consumption in all tissue - glucose absorption.
- ↑ HR, Cop, excitability & conductivity (+ arteriolar dilatation).
- ↑ skeletal growth & sexual maturation.
- ↓ serum cholesterol level.

Regulation of hormone production

- Hypothalamic TRH stimulates pituitary production of TSH, which in turn stimulates thyroid hormone synthesis and secretion.
- The presence of adequate thyroid hormone sends a negative feedback signal that inhibits TRH and TSH production.
- When the level of thyroid hormone in the serum decreases the production and release of TRH and TSH is enhanced.
- TSH, secreted by thyrotrope cells of the anterior pituitary, plays a pivotal role in control of pituitary - thyroid axis and serves as the most useful physiologic marker of thyroid hormone function.

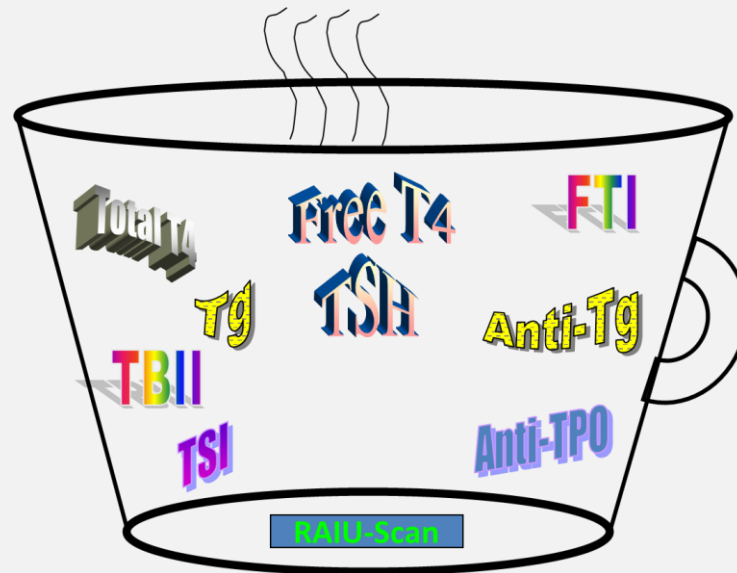


Thyroid hormone synthesis

- Iodine trapping → iodide absorbed from the intestine circulates in the blood & is picked up by the thyroid under the effect of TSH where it is converted to iodine (by the peroxidase enzyme).
- Binding → iodine + tyrosine → iodinose enz → -Mono-iodotyrosine
- Di - iodotyrosine.
- Coupling → mono - iodotyrosine + Di → iodotyrosine → - T4 (thyroxin)
- Tri-iodotyrosine (T3).
- Releasing → T4, T3 combine with globulin to form thyroglobulin
→ protease → T4, T3.
- Release of thyroid hormones is controlled by TSH which depends on the feed-back mechanism.
- Thyroid hormones in the circulation are bound to plasma proteins mainly thyroxin binding globulin (TBG) 75 %, prealbumin 15 %, albumin 10%.

MIT, monoiodotyrosine; DIT, diiodotyrosine; T3, triiodothyronine; T4, thyroxine; TG, thyroglobulin.

Thyroid Function Test



1.Measurement of : [by RIA].

- Total T4 (normal : 4 -12 ug/ dl).
- Total T3 (normol : 80-120 ng/ all).

N.B: Total T3, T4 are affected by changes in TBG.

TBG is ↑in:	TBG is ↓in:
• Pregnancy. • Oestrogen. • Congenital.	• L.C.F. • Nephrotic S • Malnutrition. • Congenital. • Androgens.

2. Measurement of free T3,T4 (0.4 ng /dl, 1.6 ng/ dl respectively).

3. Serum TSH: normal :0.5- 5 uU/ml.

- It is the most sensitive test for 1ry hypothyroidism. Thus, it can differentiate between 1ry (↑) & 2ry hypothyroidism (↓)

Primary: Thyroid disease → Hyperthyroidism → ↑ T4 + ↓ TSH.

Hypothyroidism → ↓ T4 + ↑ TSH.

Secondary: Pit. Disease → Hyperthyroidism → ↑ TSH + ↑ T4.

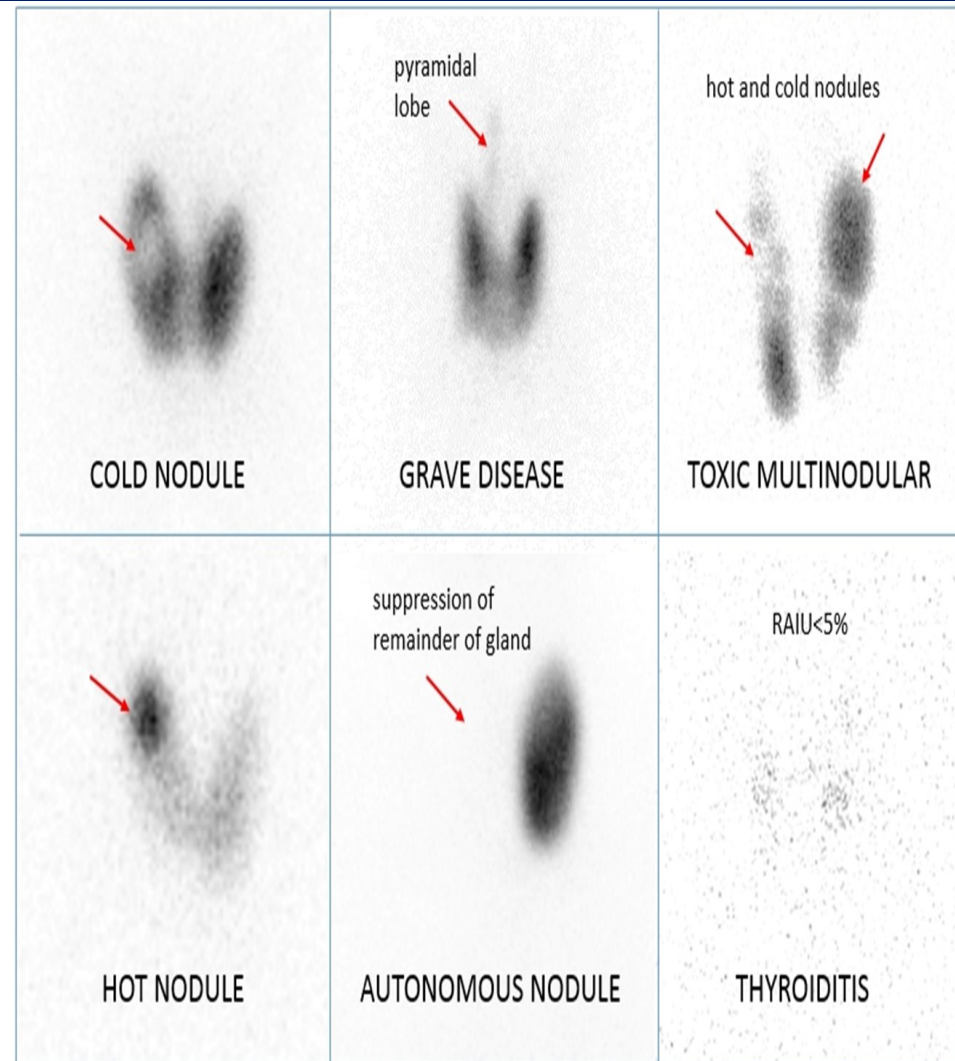
Hypothyroidism → ↓ TSH + ↓ T4.

4. Radio - active iodine uptake (R.A.I.U): I131 or I123 'preferable'

- Normal uptake (24 hours after giving the isotope) → 5 - 30 % of the taken dose.
- Useful in the diagnosis of hyperthyroidism.

N.B:

- R.A.I.U. is less useful in diagnosis of hypothyroidism because it discriminates poorly between low & normal uptake.
- Most causes of thyrotoxicosis lead to ↑ R.A.I.U except :
 - o Thyrotoxicosis factitia
 - o Thyroiditis (subacute & chronic).
 - o Ectopic thyroid tissue



⚠ Caution

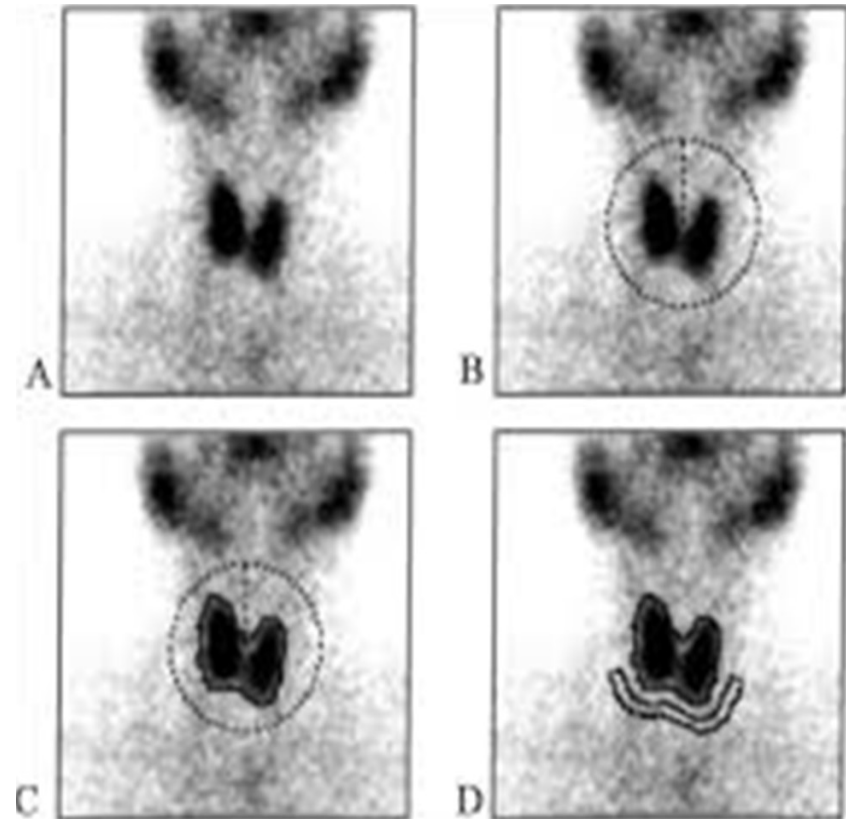
Radioactive iodine uptake scans are contraindicated during pregnancy and lactation

5. T3 suppression test :

Serum T4 or RAIU is compared before & after T3 administration → a failure of suppression is diagnostic of hyperthyroidism.

6. Thyroid scanning (using 99mTC): of value for

- Define areas of ↑ (hot nodule) or ↓ (cold nodule) uptake.
- Retrosternal goiter.
- Ectopic thyroid tissue.
- Functioning metastasis of thyroid carcinoma.



7. Anti-thyroid antibodies :

- a) Thyroid stimulating immunoglobulin, TSI (previously called long thyroid stimulator LATS) in → Graves' disease.
- b) Antimicrosomal & antithyroglobulin antibodies: in hashimoto's thyroiditis & may be in Iry hypothyroidism, Graves' disease
- c) TSH binding inhibitory immunoglobulin (TBII): in → Iry hypothyroidism.
- a) Serum thyroglobulin: ↑ in → differentiated cancer thyroid.

8. Thyroglobulin:

Useful marker to follow in differantaited thyroid cancer after thyroidectomy

	TSH receptor antibodies (TRAb)	Thyroid peroxidase antibodies (TPOAb)	Thyroglobulin antibodies (TgAb)
<u>Graves disease</u>	~ 90%	~ 70%	~ 50-70%
<u>Hashimoto thyroiditis</u>	~ 10-15%	> 90%	> 80%
<u>Thyroid cancer</u>	No association	Sporadic	~ 25% (important for follow-up!)
<u>Other conditions</u>	~ 15% in multinodular goiter	> 60% in <u>postpartum thyroiditis</u>	~ 40% in other autoimmune diseases (eg, <u>type 1 diabetes mellitus</u>)
<u>General population</u>	Negative	~ 5%	~ 5%

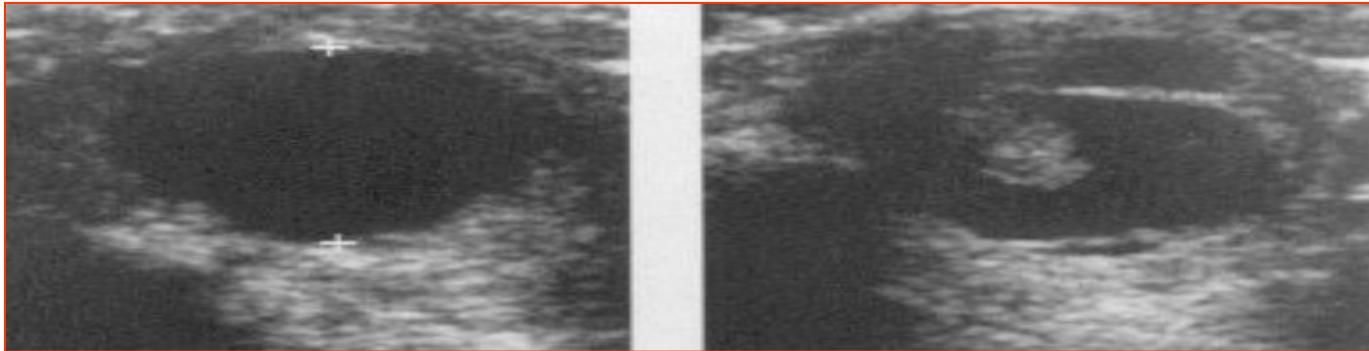
9 - Thyroid Ultrasound

Advantages

- Cystic vs. solid.
- Blood flow visualization (Doppler)
- Biopsy guidance, also of LN
- Moderate precision in volume estimation

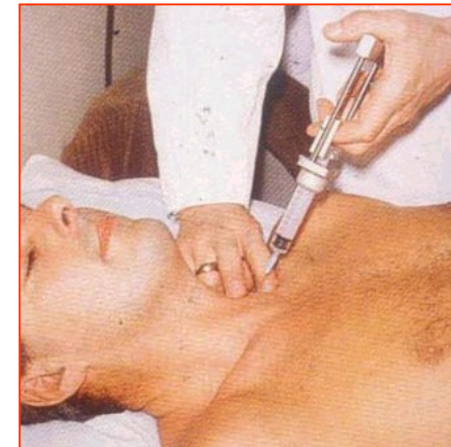
Disadvantages

- Operator dependency
- No information of functionality
- Not feasible in substernal goiter
- Poor prediction of malignancy



10- FNA biopsy

- Ideally from periphery of lesion.
- Reaspirate after fluid drawn.
- Immediately smeared and fixed.

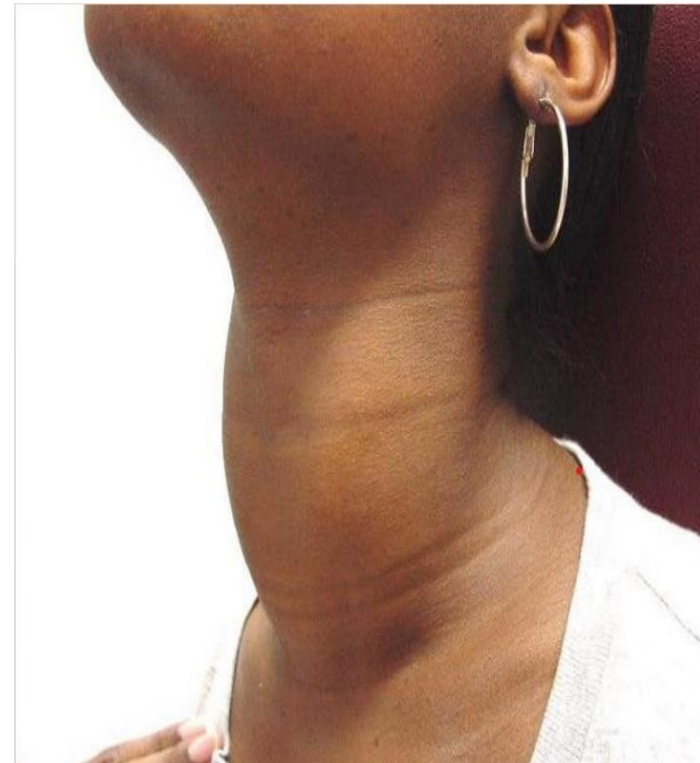




Thyroid Disorders

Goiter

- **Def:** is a diffuse or focal enlargement of the thyroid gland that may be metabolically hyperactive, hypoactive, or normoactive (Euthyroid).
- **Patients may be:**
 - **Hyperthyroid** → Graves' disease, Multinodular Goiter.
 - **Euthyroid** → Simple diffuse goiter.
 - **Hypothyroid** → Hashimoto thyroiditis.
- **Thyroid Ultrasound** is the most accurate method to assess thyroid size.





THYROTOXICOSIS

Definition:

This term refers to the clinical syndrome resulting from excess circulating thyroid hormones. Hyperthyroidism refers to cases where the thyroid gland is the source of these circulating thyroid hormones.

Causes of thyrotoxicosis :

➤ **Thyroid hyperfunction** due to:

- ✓ 1. Grave's disease. The most common cause of hyperthyroidism.
- 2. Toxic nodule: goitre.
- 3. Toxic adenoma.
- 4. iodine induced hyperthyroidism (Jod basedow phenomenon).
- 5. TSH - secreting pituitary tumor.
- 6. Choriocarcinoma (uterine or testicular), hydatidiform mole (both secrete HCG which is similar to TSH).

➤ **Abnormal thyroid hormone release** due to :

1. Subacute thyroiditis.
2. Chronic thyroiditis with transient thyrotoxicosis (hashitoxicosis)
3. Hamburger thyroiditis

➤ **Extra-thyroid. source of hormone** :

1. Thyrotoxicosis factitia. exogenous intake of thyroxin
2. Ectopic thyroid tissue
 - o Struma ovarii.
 - o Functioning metastatic thyroid carcinoma(follicular).

GRAVES' DISEASE

' Basedow disease' Toxic diffuse Goitre

Aetiology:

- It is due to the presence in the serum of abnormal thyroid stimulator called thyroid stimulating immunoglobulin [TSI] (previously termed LATS) that binds to TSH receptors (TSH-receptor antibodies (TRAb)) → diffuse goitre + ↑
- Thyroid hormone production. May be associated with other autoimmune disorders e.g. autoimmune polyglandular syndrome II
- Associated with HLA-B8, DR3. Smoking ↑ the risk.
- This stimulator is an auto-antibody belonging to IgG class (& so can cross the placenta).

Clinical features :

- A disease of middle aged ♀ (30 - 50 years). “ ♀ > ♂ by 5 10 times ”.
- The onset is insidious & the course shows remissions & exacerbations.

(The classic clinical triad is a diffuse vascular goitre, ophthalmopathy, and Dermopathy).

1.General symptoms “ metabolic ”

- Progressive weight loss inspite of good appetite.
- Intolerance to hot weather (patients tolerate winter much better than summer).

2.Neuropsychiatric manifestations :

- Nervousness, emotional liability
- Brisk reflexes.
- Fine tremors (particularly of the outstretched hands & the protruded unsupported tongue).

3. Cardio-vascular manifestations :

- Sinus tachycardia, S.V.T., AF (any arrhythmia except heart block).
- High Cop failure.
- Flow murmur.

4. gastro-intestinal manifestations :

↑ appetite & ↑ motility of GIT → diarrhea (rare).

5. Musculo-skeletal manifestations:

- Myopathy, myasthenia gravis.
- Bone resorption exceed bone formation :
 - Hypercalciuria & Sometimes hypercalcaemia.
 - Osteoporosis in elder patients.

6.Reproductive manifestations :

- in women → oligomenorrhea & ↓ fertility.
- in men → impotence, ↓ sperm count & gynaecomastia
(due to ↑ peripheral conversion of androgen to estrogen).

7.Skin manifestations:

- warm with excessive generalized sweating.
- Onycholysis (retraction of the nail from its bed), 'plummer nail'.
- Orange peel thickening of the pretibial area (pretibial myxedema) → rare
- Clubbing of the fingers & toes (thyroid acropachy).



8. Thyroid gland:

- Usually diffusely enlarged firm.
- Bruit may be heard (due to $\uparrow$ blood flow).



9.Ocular manifestations : divided into 2 components :

- **Spastic** : due to sympathetic overactivity:

(**DR must be very simple**)

- o Tremors of the closed eye lids → (Rosenbach's sign).
- o Failure of wrinkling of the forehead on upward gaze → (Joffroy's Sign).
- o Starring look → (Dalrymple Sign).
- o Lid lag → (Von Graefe's sign).
- o Infrequent blinking → (Stellwag Sign).

- **Mechanical** :

Due to inflammatory changes in the contents of the orbital cavity due to autoimmune process & thus occur only in Graves' disease (infiltrative ophthalmopathy).

- o Proptosis, ophthalmoplegia (diplopia).
- o Inability to converge eyes at near Vision "mobius Sign".
- o Conjunctivitis, chemosis, peri-orbital swelling. Later on, corneal ulceration, optic neuritis & optic atrophy may occur.

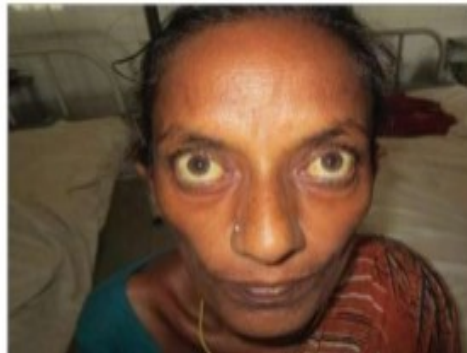
Von Graefe's sign

- The upper eye lid lags the eyeball as the patient is asked to look downwards.

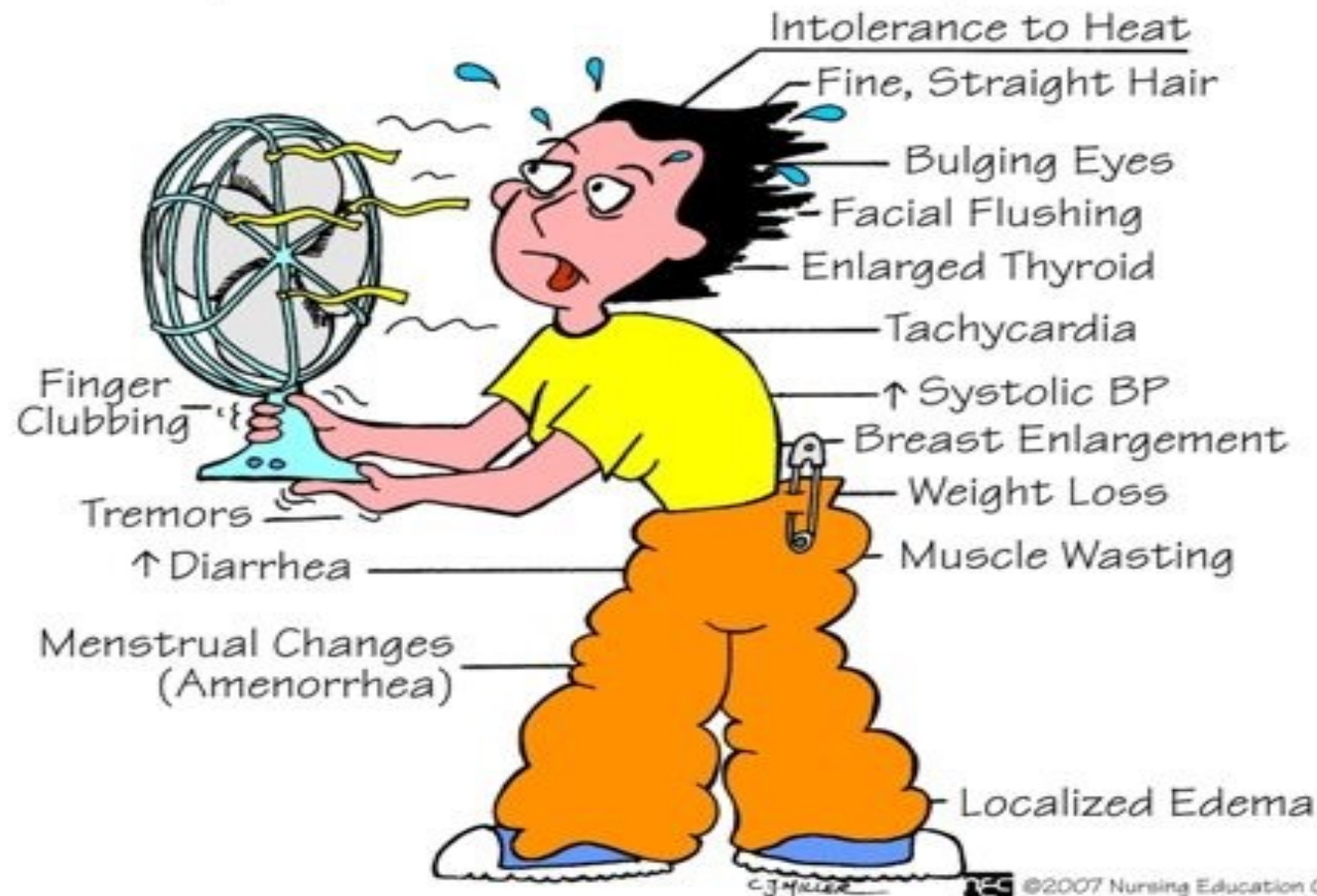


Joffroy's sign

- Absence of wrinkling on the forehead when the patient looks upwards with the face inclined downwards



HYPERTHYROIDISM



Clinical triad of Graves disease: diffuse goiter, ophthalmopathy, and pretibial myxedema

10. thyrotoxicosis crisis “ thyroid storm ” :

- **Def:** It is a dangerous condition of decompensated thyrotoxicosis.
- **Causes :**
 - Excessive manipulation of the thyroid during thyroidectomy.
 - Neglected severe hyperthyroidism + intercurrent illness e.g. Pneumonia, gastroenteritis.
- **Clinical picture :**
 - Tachycardia.
 - Fever.
 - Irritability.
 - Psychosis.
 - Nausea.
 - Vomiting.
 - Diarrhea.

Investigations: → see evaluation of the thyroid functions.

Differential diagnosis:

1. From similar conditions:

- Anxiety neurosis: (cold sweaty Skin + normal sleeping pulse).
- Anemia.
- Cardiac failure. . COPD . Malignancy.
- Malabsorption syndrome & DM . Myopathies & myasthenia gravis.

2. From other causes of thyrotoxicosis:

Nodular thyroid gland (+ ↑ RAIU) → toxic nodular goiter or toxic adenoma.

- Tenderness of the thyroid & firm nodularity → subacute thyroiditis.
- A small firm non tender goitre → chronic thyroiditis (hashimoto thyroiditis → hashitoxicosis).
- Absence of a palpable thyroid gland → extra thyroid source of the hormone e.g: Struma ovarii, Thyrotoxicosis factitia.

	Graves's disease (1ry thyrotoxicosis) Toxic diffuse goitre	Toxic nodular goiter
Thyroid gland	Diffusely enlarged	Nodular
Cvs manifestation	+	+++
Nervous manifestation	+++	+
Infiltrative ophthalmoathy	Usually present	abscent
Treatment	Usually medical	Usually surgical

N.B: Antithyroid antibodies can help in the differentiation.

Treatment

1. Medical :

➤ *Indicated in :*

- thyrotoxicosis in pregnant ♀
- Cases complicated by HF.
- Graves' disease in a patient < 25 years
- Pre-medication before operation.

➤ *Contraindicated in :*

- Huge goitre → pressure symptoms
- Retrosternal
- Suspicion of malignancy

A. Thionamide drugs

➤ *Mechanism of action:*

- Inhibit thyroid peroxidase → ↓ synthesis of thyroid hormones.
- Propyl thiourasil → ↓ peripheral production of T3 from T4.
- Carbimazol → ↓ production of TSI (LATS) 'immuno-suppressive effect'.

➤ *Dose :*

- Propyl thiouracil: 300 - 600 mg / day (in 3 divided doses).
- Methimazol: 30 - 60 mg / day (once daily or in 3 divided doses).
- Carbimazol (Neomercazol): 30 - 60 mg / day (once daily or in 3 divided doses).

- *Follow up:* → the dose can be decreased to $\frac{1}{2}$ - $\frac{2}{3}$ the initial dose (after control of the disease). Therapy should be maintained for 1 - 2 years to induce long term remission.
- *Side effects :*
 - Agranulocytosis (do blood examination on feeling sore throat).
 - Arthralgia.
 - Skin rash.
 - Serum sickness.
- *Prognosis:*
 - Long term remission (when course is completed) in $\frac{1}{2}$ of patients
 - Relapses usually occur within the 1st year after stopping therapy
 - Some patients develop hypothyroidism after a long time after treatment

B. Beta adrenergic blockers

- Propranolol (Inderal) is used.
- Mechanism of action :
 - ↓ Excessive adrenergic activity.
 - ↓ Conversion of T4 to T3.
- Dose : → 20- 40 mg / 6 hrs (adjusted to ↓ HR to about 80 /m)

C. Other agents

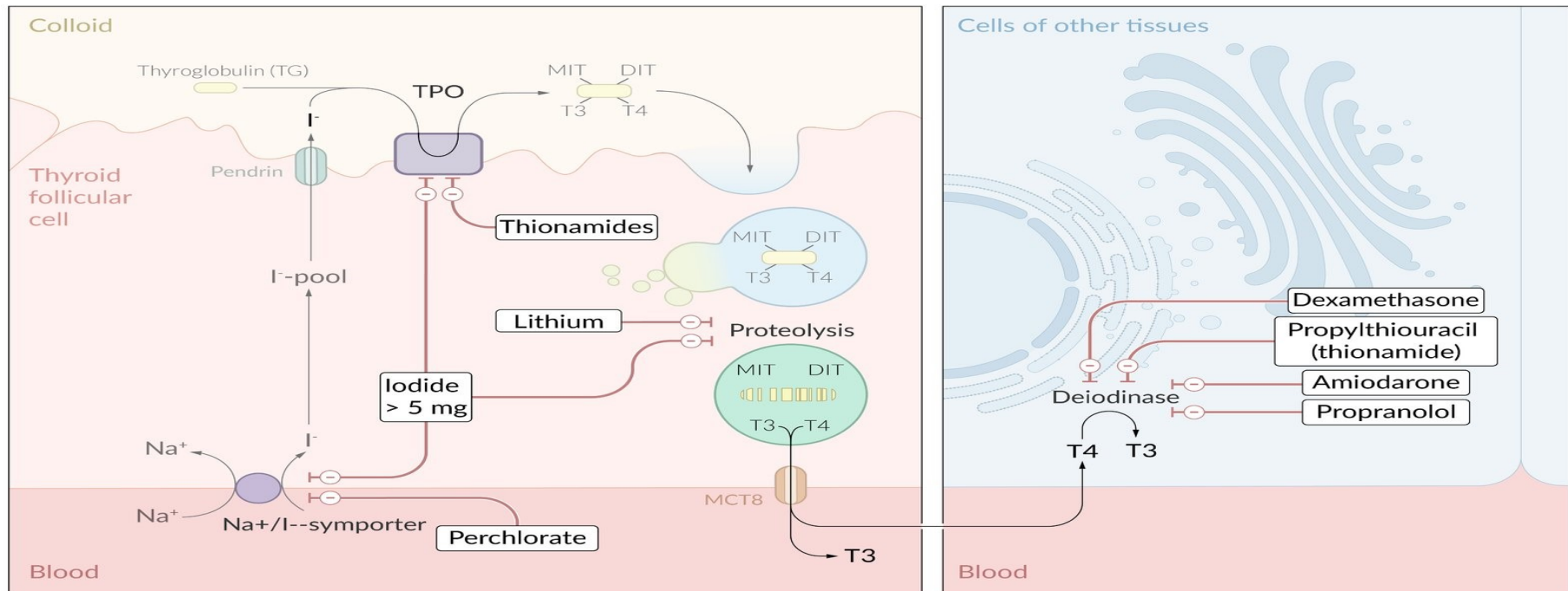
1. Na ipodate (oragraphin): 1 gm / day → ↓ T 4 to T3.
2. K. iodine →
 - Mainly used to prepare patient for surgery (by ↓ gland vascularity).
 - 5 drops (250 mg) bid.
 - But escape from its effect occurs after 10 days.
3. Dexamethasone: 8 mg / day.
 - ↓ Secretion of thyroid hormone & ↓ T4 → T3.

⚠ Caution

Obtain a CBC and liver chemistries immediately if patients develop fever or signs of hepatotoxicity while taking ATD. 18

💡 Clinical pearl

ATDs are ineffective in the management of destructive thyroiditis-induced thyrotoxicosis, which is caused by a release of preformed thyroid hormones by the damaged follicles. 18



2. Radio-iodine 131 therapy:

➤ Indicated in:

- Recurrence after thyroidectomy
- After failure of medical ttt & the patient is unfit for surgery.
- Patients in whom surgery is indicated but refusing the operation

➤ Contraindicated in:

- During pregnancy, lactation & childhood
- Huge & retrosternal goiter.

➤ *Side effects :*

- Hypothyroidism :
 - o If it occurs within the 1st month after ttt → transient.
 - o If occurs after one year → permanent.
- May cause thyroid carcinoma.
- Foetal abnormalities & hypothyroidism of the new born if given by mistake during pregnancy.

N.B:

- Some patients may Show acute release of the hormone on initiation of the ttt so, thionamide drugs have to be given before I131 & stopped 3-5 days before giving 1131 in order not to interfere with the uptake of iodine" (to be restarted 14 days later).
- Antithyroid drugs may be given for next few months since radio-iodine doesn't achieve its effect till then.

3.Surgical subtotal thyroidectomy : (or total)

➤ *Indications :*

- Allergy to Antithyroid drugs.
- Patients refusing I 131
- Big & retrosternal goiter.
- Multinodular toxic goiter.
- Malignant suspicion.

➤ *Preparation :*

- Treatment with thionamide drugs for several months.
- Inorganic iodine 7 - 10 days before the operation → to ↓ vascularity.
- Beta-blockers may be used for short periods.

➤ *Complications :*

- Hypothyroidism.
- Hypoparathyroidisin
- Recurrent hyperthyroidiam.
- Recurrent laryngeal nerve injury.

4. Treatment of pregnant ♀

- a) I^{131} is contra-indicated because it can cross the placenta.
- a) Propyl-thiouracil crosses the placenta to lesser extent than the other thionamide preparations, so it is the preferable drug.
- b) if necessary → surgical thyroidectomy → best performed in the 1st or 2nd trimester because general anesthesia in the 3rd trimester may lead to premature labour.
- c) The newborn: may develop hyperthyroidism due to transplacental transmission of TSI so, special attention must be paid for him.

5- Treatment of complications :

Ocular complications :

- Protect the eye with ointment, sunglasses, and methylcellulose.
- Prednisone ↓ lymphocytic infiltration.
- Orbital decompression.

Heart failure :

- Bed rest, salt restriction, digitalis & diuretics.

Pretibial myxoedema :

- Remits spontaneously after months or years.
- . Local Betamethasone cream (local steroid) , relieves pruritis.

6.Treatment of thyrotoxic crisis :

- Ice bags (to control hyperpyrexia), fluids & electrolytes.
- IV. Dexamethasone 4 - 8 mg / day (3 unless contraindication by severe infection). Decrease conversion and treat underlying adrenal insufficiency
- b- blockers: eg, IV. propranolol to ↓ H.R.
- Ipodate (Na ipodate) 1 gm / day for 2 weeks.
- Antithyroid in a large dose : propyl thiouracil 600 mg then 300 mg/ 6 h can be used but methimazole is the treatment of choice because of SE of PTU.
- Antibiotics → to control an underlying infection.

⚠ Caution

If thyroid storm has led to congestive heart failure, esmolol is the preferred beta blocker. All patients receiving beta blockers should be monitored carefully for signs of heart failure.

💡 Student pearl

Treat thyroid storm with PROverbial PROficiency and POetic GLUttony: PROpranolol, PROpylthiouracil, POtassium iodide, and GLUcocorticoids.

Case scenario :

A 35-year-old woman comes to the physician because of intermittent episodes of heart palpitations for 3 weeks. The palpitations are accompanied by chest pain that is sharp and localized to the left 3rd intercostal space. The patient does not smoke. She drinks one to two glasses of wine on the weekends. She appears nervous. Her pulse is 110/min and irregularly irregular, and blood pressure is 135/85 mmHg. Examination shows a fine tremor on both hands and digital swelling. There is retraction of the left upper eyelid. A systolic ejection murmur is heard along the right upper sternal border. The extremities are warm, and pulses are 2+ bilaterally.

Which of the following is the most appropriate pharmacotherapy to address this patient's symptoms?

- A. Propylthiouracil
- B. Propranolol
- C. Digoxin
- D. Amiodarone
- E. Warfarin
- F. Diltiazem

Summary

I just need
the main ideas



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Thank You